Day 7 — Dockerfile Theory

DEVOPS-007: Dockerfile & Custom Docker Image | Task 1: Understand Dockerfile | Time: 45 Minutes

1. What is a Dockerfile?

A Dockerfile is a plain text file containing a set of step-by-step instructions that Docker reads to automatically build a custom image. Each line is an instruction (like FROM, COPY, RUN) that tells Docker what base to start from, what files to add, what commands to run, and how the resulting container should behave when it starts.

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2. Why do we need a Dockerfile?

Without a Dockerfile, building an image means manually configuring a container step by step and committing it — which isn't repeatable or shareable. A Dockerfile makes image creation automated, version-controlled, and consistent: anyone can rebuild the exact same image just by running docker build against the same file. It's essentially "infrastructure as code" for container images.

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3. What is FROM?

FROM sets the base image that the new image is built on top of. It's almost always the very first instruction in a Dockerfile.

```
FROM ubuntu:22.04
```

4. What is WORKDIR?

WORKDIR sets the working directory inside the container for any instructions that follow (RUN, COPY, CMD, etc.). If the directory doesn't exist, Docker creates it automatically. It saves having to write full paths repeatedly.

```
WORKDIR /app
```

5. What is COPY?

COPY copies files or folders from the host machine (build context) into the image's filesystem at a specified destination path.

```
COPY . /app
```

6. What is RUN?

RUN executes a command at build time and bakes the result into a new image layer — typically used for installing packages or dependencies. It only runs once, while the image is being built.

```
RUN apt-get update && apt-get install -y curl
```

7. What is CMD?

CMD specifies the default command that runs when a container starts from the image. It can be overridden at runtime if a different command is passed to docker run. A Dockerfile can only have one

effective CMD.

```
CMD ["nginx", "-g", "daemon off;"]
```

8. What is EXPOSE?

EXPOSE documents which port(s) the container listens on at runtime. It's mainly informational/metadata — it doesn't actually publish the port to the host. You still need -p when running the container to map it to a host port.

```
EXPOSE 80
```

9. Difference between RUN and CMD?

RUN executes during the image build process and creates a new layer — it's used to set up the environment (installing software, creating directories, etc.). CMD only defines what command runs when a container starts from the finished image, and doesn't execute at build time. A Dockerfile can have many RUN instructions but effectively only one CMD (the last one wins if multiple are specified).

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10. Difference between CMD and ENTRYPOINT?

CMD provides default arguments/command that can be completely overridden by whatever is passed after the image name in docker run. ENTRYPOINT defines the main command that always runs and is much harder to override (you need --entrypoint to replace it) — anything passed in docker run is instead appended as arguments to it. They're often used together: ENTRYPOINT sets the fixed executable, and CMD supplies default arguments to it that a user can still override.